

# A267 – The Stipulation $\alpha = 1$ : Charge Redefinition and Conversion Examples

Johann Pascher

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## Contents

|     |                                                                           |   |
|-----|---------------------------------------------------------------------------|---|
| .1  | Purpose . . . . .                                                         | 1 |
| .2  | Stipulation: Heaviside–Lorentz with $\alpha = 1$ . . . . .                | 1 |
| .3  | Back-Computation as Substitution . . . . .                                | 1 |
| .4  | Conversion Examples . . . . .                                             | 1 |
| .5  | The 137-Ladder: What $\alpha = 1$ Makes Visible . . . . .                 | 2 |
| .6  | Why This Stipulation – Comparison with Alternatives . . . . .             | 2 |
| .7  | FFGFT Connection: Back-Computing $\alpha$ Means Inserting $\xi$ . . . . . | 2 |
| .8  | Symbol Glossary . . . . .                                                 | 3 |
| .9  | Verification Script . . . . .                                             | 3 |
| .10 | Open Questions . . . . .                                                  | 3 |
| .11 | Supersedes . . . . .                                                      | 3 |
| .12 | Use of AI Tools . . . . .                                                 | 3 |

## .1 Purpose

**[STIPULATION]** In FFGFT  $\alpha$  follows from the geometry (A130). This document treats the complementary question: how does one set  $\alpha = 1$  *as a computational tool* in a way that is fully reversible and makes the coupling structure visible? The answer is the Heaviside–Lorentz stipulation  $e^2 = 4\pi\alpha$  combined with  $\alpha = 1$  – not a unit choice but a *stipulation* with preserved trace.

## .2 Stipulation: Heaviside–Lorentz with $\alpha = 1$

**[STIPULATION]** In natural units ( $\hbar = c = 1$ ) and the Heaviside–Lorentz system, the *defining equation* is

$$e^2 = 4\pi\alpha.$$

The fine-structure constant is set to  $\alpha = 1$ . This is not a *norm* (unit choice) – a dimensionless number is invariant under rescaling of base units – but a *stipulation*: the coupling is adopted as a computational unit.

**[B]** The charge carries the  $4\pi$ ; what is set to 1 is  $\alpha$ , not  $e$ . From  $e^2 = 4\pi\alpha$  with  $\alpha = 1$  follows the value  $e = \sqrt{4\pi} \approx 3.545$  – not  $e = 1$ . The relation  $e^2 = 4\pi\alpha$  remains as a defining equation and cannot be dissolved by the stipulation.

### .3 Back-Computation as Substitution

**[B]** Because  $e^2 = 4\pi\alpha$  remains as a relation, the coupling power is automatically carried along. Back-computation is pure substitution

$$\alpha \longrightarrow \frac{1}{137.036},$$

applied to the  $\alpha$ -order of the process:

- per vertex a factor  $e = \sqrt{4\pi\alpha}$ ; compared to the  $\alpha=1$  value  $\sqrt{4\pi}$ , that is a  $\sqrt{\alpha}$  back,
- amplitude with  $V$  vertices  $\propto (4\pi\alpha)^{V/2} \Rightarrow \times\alpha^{V/2}$ ,
- rate/cross section  $\propto (4\pi\alpha)^V \Rightarrow \times\alpha^V$ ,
- compact: process of rate order  $\alpha^n \Rightarrow \times\alpha^n$ .

**[B]** The  $4\pi$  is not double-counted: it is already in  $e^2$  and in the phase-space normalisation. Only  $\alpha^n$  is reset, not  $(4\pi\alpha)^n$ .

### .4 Conversion Examples

**[K]** All quantities in  $\hbar = c = 1$ , Heaviside–Lorentz. Back-factor = the  $\alpha$ -order of the expression.

| Quantity               | Physical form                             | With $\alpha=1$        | Back-factor         |
|------------------------|-------------------------------------------|------------------------|---------------------|
| Coulomb potential      | $V = \alpha/r$                            | $1/r$                  | $\times\alpha$      |
| Class. electron radius | $r_e = \alpha/m_e$                        | $1/m_e$                | $\times\alpha$      |
| Compton wavelength     | $\lambda_C = 1/m_e$                       | $1/m_e$                | $\times 1$          |
| Bohr radius            | $a_0 = 1/(\alpha m_e)$                    | $1/m_e$                | $\times\alpha^{-1}$ |
| Rydberg / binding      | $E = \frac{1}{2}\alpha^2 m_e$             | $\frac{1}{2}m_e$       | $\times\alpha^2$    |
| Thomson cross section  | $\sigma_T = \frac{8\pi}{3}\alpha^2/m_e^2$ | $\frac{8\pi}{3}/m_e^2$ | $\times\alpha^2$    |

### .5 The 137-Ladder: What $\alpha = 1$ Makes Visible

**[B]** The three atomic length scales stand in the ratio

$$r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}.$$

At  $\alpha = 1$  they all collapse to  $1/m_e$ : they are not three independent scales but *one* scale with three  $\alpha$ -powers.  $a_0$  is exactly 137× larger than  $\lambda_C$ , and  $\lambda_C$  exactly 137× larger than  $r_e$ .

**[B]** With  $\alpha = 1/137$  written out everywhere, this common origin remains hidden – three different awkward numbers appear and the  $\alpha$ -power as ordering principle is not visible. The  $\alpha = 1$  stipulation reveals it.

### .6 Why This Stipulation – Comparison with Alternatives

**[S] Versus Gaussian ( $e^2 = \alpha$ ).** Also works, but places the  $4\pi$  in the field equations rather than in Coulomb's law. Heaviside–Lorentz places the  $4\pi$  where they geometrically belong (sphere surface, Coulomb) and leaves the Maxwell equations  $4\pi$ -free.  $\alpha$  is then the pure coupling, not a mixture of coupling and geometry.

**[S] Versus  $c = \hbar = 1$  unit choice.** The setting  $c = \hbar = 1$  destroys the powers – the trace is gone, back-computation runs only via dimensional analysis (A266). The defining equation  $e^2 = 4\pi\alpha$ , by contrast, *retains the trace*: the coupling power is carried along because the relation stands. Stipulation with preserved trace, not norm with erased trace – hence constructively reversible, not merely justifiably reconstructable.

## .7 FFGFT Connection: Back-Computing $\alpha$ Means Inserting $\xi$

**[B]**  $\alpha$  is in the FFGFT corpus not a free input but follows from the geometry (A130):

$$\alpha^{-1} = \frac{K_{\text{frak}}}{\xi m_e m_\mu} = 137.06 \quad \text{from} \quad \xi = \frac{4}{30000}, \quad D_f = 3 - \xi, \quad E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}}.$$

The stipulation  $\alpha = 1$  isolates precisely the point at which the  $\xi$ -geometry enters. Every reset  $\alpha^n$  is a location at which  $D_f = 3 - \xi$  becomes visible.

**[B]** Within the corpus, “back-computing  $\alpha$ ” therefore does not mean “inserting  $1/137$ ” but “inserting the  $\xi$ -structure” – and the  $\alpha=1$  system makes countable at how many places and in which power that occurs.

## .8 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

| Symbol                          | Meaning                                      |
|---------------------------------|----------------------------------------------|
| $e^2 = 4\pi\alpha$              | Heaviside–Lorentz defining equation          |
| $e = \sqrt{4\pi} \approx 3.545$ | Elementary charge at $\alpha = 1$            |
| $r_e = \alpha/m_e$              | Classical electron radius                    |
| $\lambda_C = 1/m_e$             | Compton wavelength                           |
| $a_0 = 1/(\alpha m_e)$          | Bohr radius                                  |
| $V$                             | Number of Feynman vertices in the process    |
| $\alpha^n$                      | $\alpha$ -order / back-factor of the process |

## .9 Verification Script

- Numerical verification of all table rows with  $\alpha = 1/137.036$  and  $m_e = 0.511 \text{ MeV}$ ; ratio  $r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}$ : `python/A_Serie_Skripte/a267_alpha_eins.py`

## .10 Open Questions

**The stipulation  $\alpha = 1$  is fully worked out only for electromagnetic processes.** For weak and strong coupling ( $\alpha_W, \alpha_s$ ) an analogous table would be possible but has not been worked out.

**The connection  $\alpha^n$  as locus of  $D_f = 3 - \xi$**  is plausibly shown for simple QED processes but not for multi-loop or non-abelian processes.

## .11 Supersedes

This document subsumes: Doc. 011 (individual formulas  $r_e$ ,  $\lambda_C$ ,  $a_0$ , dimensional analysis  $\alpha = r_e/\lambda_C$ , SI numerical values of atomic scales), Doc. 015 (Heaviside–Lorentz line  $e^2/(4\pi) = 1$  in the unit systematics). The coherent conversion table with explicit  $\alpha^n$  back-factors and the 137-ladder  $r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}$  as a structural statement are not present as a unit in the legacy corpus and are newly systematised here. Related: A130 (geometric derivation  $\alpha$  from  $\xi$ ), A266 (SI back-computation).

## .12 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.